# Practical no 6

**Aim:**

To apply project scheduling and tracking techniques (CPM, PERT, and Gantt Chart) to analyze and manage the development of a construction business website.

**Write-up :**

What is project scheduling? Why is it important in software project management?

Explain CPM,PERT and Gantt chart in project scheduling.

## Objectives:

1. To break down website development into well-defined activities.
2. To calculate project completion time using **PERT**.
3. To identify the **Critical Path** using **CPM**.
4. To prepare a **Gantt chart** for project visualization and tracking.

## Step 1: Define Activities & Dependencies

| Code | Activity | Predecessor(s) |
|---|---|---|
| A | Requirement Gathering & Analysis | – |
| B | System & Website Design | A |
| C | Database Design | B |
| D | Frontend Development (UI/UX) | B |
| E | Backend Development (Server + APIs) | C |
| F | Integration (Frontend + Backend + DB) | D, E |
| G | Testing & Debugging | F |
| H | Deployment & Maintenance Plan | G |

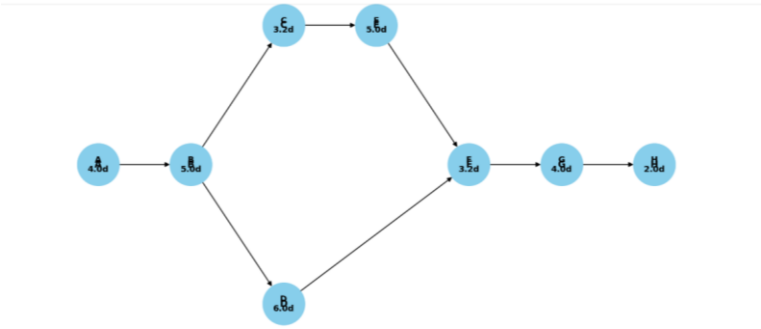
## Step 2: Time Estimation (Optimistic, Most Likely, Pessimistic)

Using **PERT formula:**

$$TE = \frac{O + 4M + P}{6}$$

| Activity | O (days) | M (days) | P (days) | TE (days) |
|---|---|---|---|---|
| A | 2 | 4 | 6 | 4.0 |
| B | 3 | 5 | 7 | 5.0 |
| C | 2 | 3 | 5 | 3.2 |
| D | 4 | 6 | 8 | 6.0 |
| E | 3 | 5 | 7 | 5.0 |
| F | 2 | 3 | 5 | 3.2 |
| G | 2 | 4 | 6 | 4.0 |
| H | 1 | 2 | 3 | 2.0 |

## Step 3: Draw PERT / CPM Network Diagram



**A → B → {C, D} → E → F → G → H**

- **Critical Path:** A → B → D → F → G → H
- **Project Completion Time: 24.2 days ≈ 24 days**

## Step 4: Gantt Chart (Timeline Representation)

| Activity | Duration (days) | Start | End |
|---|---|---|---|
| A | 4 | Day 1 | Day 4 |
| B | 5 | Day 5 | Day 9 |
| C | 3.2 | Day 10 | Day 13 |
| D | 6 | Day 10 | Day 15 |
| E | 5 | Day 14 | Day 18 |
| F | 3.2 | Day 19 | Day 21 |
| G | 4 | Day 22 | Day 25 |

| Activity | Duration (days) | Start | End |
|---|---|---|---|
| H | 2 | Day 26 | Day 27 |

